

Letter II

The Obidi Action, the Obidi Field Equations, and the Mathematical Architecture of Entropic Emergence

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"The physicist who is interested in the logical development of a physical theory is satisfied only when the theory is expressed in mathematical form, and this mathematical form is the real vehicle of thought."

— *Paul Adrien Maurice Dirac*,

The Principles of Quantum Mechanics

(1930)

Abstract

This second installment of the *Living Review Letters* undertakes a systematic exposition of the mathematical architecture underpinning the Theory of Entropicity (ToE). Where Letter I established the ontological foundations — the primacy of the entropic field $S(x)$, the No-Rush Theorem, the Entropic Time Limit (ETL), and the philosophical departure from geometry-first physics — this Letter II turns to the formal variational and field-theoretic machinery that gives the theory its predictive and unifying power. The scope of the present letter is comprehensive. We examine in detail: (i) the dual structure of the Obidi Action, comprising the Local Obidi Action (LOA) and the Spectral Obidi Action (SOA), which together define the variational foundation of the theory; (ii) the information-geometric underpinnings through the Fisher-Rao metric, the Fubini-Study metric, and the Amari-Čencov alpha-connection formalism, which supply the rigorous geometric substrate upon which the entropic field propagates; (iii) the derivation and structure of the Obidi Field Equations (OFE), also known as the Master Entropic Equation (MEE), which constitute the central dynamical equations of the theory; (iv) the Obidi Curvature Invariant (OCI) of $\ln 2$ as the fundamental quantum of distinguishable curvature — a prediction unique to the Theory of Entropicity (ToE); (v) the Vuli-Ndlela Integral as an entropy-constrained reformulation of Feynman's path integral, introducing irreversibility directly into quantum amplitudes; (vi) the reproduction of Einstein's field equations of General Relativity as a limiting case of the Obidi Field Equations, including classical tests such as the perihelion precession of Mercury; (vii) the Generalized Entropic Expansion Equation (GEEE) and its implications for entropic cosmology without the postulation of dark energy; and (viii) the mathematical subsumption of Ginestra Bianconi's "Gravity from Entropy" (GfE) framework, published in *Physical Review D* (2025), as a special instance within the broader architecture of the Theory of Entropicity (ToE). A critical assessment of open mathematical challenges and future

directions is also provided. Throughout, the aim of this review is to evaluate the mathematical coherence, scope, and current limitations of the Theory of Entropicity (ToE) with the rigor expected of a living review in theoretical physics.

1. Preamble and Motivation — From Ontology to Architecture

In Letter I of this series, we undertook an extensive review of the ontological revolution proposed by the Theory of Entropicity (ToE). The central claim was striking in its simplicity and far-reaching in its consequences: entropy is not a derived statistical quantity, not a bookkeeping device for counting microstates, not a secondary descriptor appended to a more fundamental dynamical substrate — but rather the fundamental dynamical field, $S(x)$, from which spacetime, geometry, forces, matter, and information emerge. We reviewed the philosophical scaffolding of this claim: the doctrine of ontodynamics, which asserts that entropy is the sole ontological primitive; the No-Rush Theorem, which establishes that no physical process can unfold faster than the rate at which entropic distinctions can be established; the Entropic Time Limit (ETL), which quantifies the minimum temporal interval required for the creation of an entangled state; and the systematic rejection of instantaneity as a physical idealization. These were the conceptual foundations — the *what* of the Theory of Entropicity (ToE).

Now, in Letter II, the focus shifts decisively from ontology to architecture, from philosophical assertion to mathematical formulation, from "what the theory claims" to "how the theory works." The central question that animates this letter is both precise and demanding: *What is the mathematical machinery that translates the ontological claim of entropic primacy into a predictive, falsifiable physical theory?*

This is the question that separates programmatic declarations from genuine theoretical physics. The history of physics teaches us, with unforgiving clarity, that an idea — however beautiful, however philosophically compelling — acquires the status of a physical theory only when it is expressed through a well-defined mathematical framework capable of generating quantitative predictions that can be compared with experiment. The transition from natural philosophy to physics, from Aristotle to Newton, was precisely this transition: the transition from qualitative description to mathematical law.

It is therefore of paramount importance that we situate the mathematical ambitions of the Theory of Entropicity (ToE) within the historical tradition of variational principles in physics. The principle of least action — or more precisely, the principle of stationary action — stands as perhaps the most powerful organizing principle in all of theoretical physics. From the Euler-Lagrange equations of classical mechanics, through the Einstein-Hilbert action of General Relativity, to the Dirac action and the Standard Model Lagrangian, every successful fundamental theory in modern physics is built upon a well-defined action functional whose variation yields the dynamical equations of the theory. The action principle is not merely a convenient mathematical device; it encodes the deepest structural features of a physical theory — its symmetries, its conservation laws (via Noether's theorem), its degrees of freedom, and its coupling structure.

The Theory of Entropicity (ToE) follows this venerable tradition by introducing the Obidi Action as its foundational variational principle. But the Obidi Action differs from its predecessors in a critical and profound respect: it is defined not on a pre-existing spacetime manifold equipped with a fixed background metric, but on an entropic manifold whose geometry is itself generated by the entropy field $S(x)$. The metric tensor $g_{ab}(S)$ is not given *a priori*; it is induced by the field whose dynamics it governs. This creates a self-referential, bootstrap structure that has no direct analog in the standard formulations of General Relativity or quantum field theory — though it

resonates, in certain structural aspects, with the background-independent program of loop quantum gravity and the Wheeler-DeWitt formulation of canonical quantum gravity.

This background-free, self-referential character places the Obidi Action in a unique position among action principles in theoretical physics. It is this feature that allows the Theory of Entropicity (ToE) to claim — in principle — the derivation of spacetime geometry as an output of the theory rather than an input. If substantiated in full mathematical detail, this would represent a qualitative advance over both General Relativity (which presupposes a differentiable manifold) and quantum field theory (which presupposes a fixed background spacetime, at least in its perturbative formulations).

The present Letter will systematically build the mathematical architecture from the ground up, following the logical chain: action principle $\rightarrow$ field equations $\rightarrow$ curvature invariants $\rightarrow$ path integrals $\rightarrow$ cosmological equations $\rightarrow$ subsumption of competing frameworks. At each stage, we will assess the mathematical rigor, the structural coherence, and the current limitations of the formalism. The aim is not advocacy but evaluation — the kind of rigorous, dispassionate analysis that the physics community requires before according a new framework the status it claims.

Let us proceed, then, from ontology to architecture.

2. The Dual Structure of the Obidi Action

The mathematical backbone of any physical theory is its action principle. The Theory of Entropicity (ToE) is founded upon a variational principle of notable structural richness: the Obidi Action. Unlike the action principles of classical mechanics, General Relativity, or the Standard Model — each of which is defined by a single action functional — the Obidi Action possesses a dual

structure, comprising two complementary components: the Local Obidi Action (LOA) and the Spectral Obidi Action (SOA). Together, these two actions define a variational architecture that is, to the best of this author's knowledge, unique in the landscape of theoretical physics.

2.1 The Local Obidi Action (LOA)

The Local Obidi Action governs the differential, local, point-by-point dynamics of the entropy field $S(x)$ on the entropic manifold. It is the component of the Obidi Action that most closely resembles the action principles familiar from classical field theory, and it is the engine that generates the Obidi Field Equations through the standard variational procedure.

The LOA contains three key structural components, each of which plays a distinct and physically motivated role:

(a) The kinetic term $A(S)(\nabla S)^2$ — This term endows the entropic field with propagation dynamics. It is structurally analogous to the kinetic term in scalar field theories, such as the Klein-Gordon field, where the term $\frac{1}{2}(\partial_\mu\varphi)(\partial^\mu\varphi)$ governs the propagation of a scalar field φ through spacetime. In the Theory of Entropicity (ToE), however, the gradient ∇S captures how entropy varies across the entropic manifold, and the coefficient $A(S)$ — a functional of the field itself — introduces a field-dependent kinetic coupling that is absent in simple scalar field theories. This field-dependence is a direct consequence of the self-referential structure of the theory: the "speed" at which entropy propagates depends on the local value of the entropy field itself.

(b) The self-interaction potential $V(S)$ — This potential describes how the entropy field interacts with itself, governing stability, phase transitions, and equilibrium configurations within the entropic manifold. The role of $V(S)$ is structurally analogous to the Higgs potential in the Standard Model — the celebrated "Mexican hat" potential $V(\varphi) = \lambda(\varphi^2 - v^2)^2$ whose spontaneous symmetry breaking gives rise to mass generation. In the Theory of Entropicity

(ToE), the self-interaction potential $V(S)$ governs the emergence of stable entropic configurations — the "ground states" of the entropic field — and the transitions between them. The precise functional form of $V(S)$ remains under active development, a point to which we shall return in the critical assessment.

(c) The matter coupling term $\eta S T^{\mu\nu}$ — This term encodes the direct coupling between the entropy field and the stress-energy tensor of matter. The coupling constant η plays a role analogous to Newton's gravitational constant G in General Relativity, but here it mediates entropic-matter interactions rather than geometric-matter interactions. Through this coupling, the entropic field "feels" the distribution of matter and energy, and matter in turn responds to the entropic gradients generated by the field. This is the term that allows the Theory of Entropicity (ToE) to make contact with gravitational phenomenology.

In its compact variational form, the Obidi Action can be written as:

$$A[S] = \int_M R(S) \sqrt{|g(S)|} d^n x$$

where $R(S)$ is the entropic curvature scalar derived from the entropy field, and the entropic metric is given by:

$$g_{ab}(S) = \nabla_a S \nabla_b S + \lambda h_{ab}$$

Here, h_{ab} is a background informational metric ensuring non-degeneracy (so that the metric remains invertible even at critical points of the entropy field where $\nabla S = 0$), and λ is a scaling parameter. The application of the principle of stationary action, $\delta A[S] = 0$, to the LOA yields the Obidi Field Equations, which we shall examine in detail in Section 4.

2.2 The Spectral Obidi Action (SOA)

While the LOA governs local, differential dynamics, the Spectral Obidi Action addresses a complementary but equally essential requirement: global, spectral, and non-local consistency. The SOA ensures that the local solutions

obtained from the LOA are globally consistent across the entire entropic manifold — that no local solution violates the global topological and spectral constraints imposed by the manifold's structure.

The SOA incorporates the spectral decomposition of the entropic field on the manifold, ensuring coherence across all scales. This is not merely a mathematical nicety; it is a physical necessity. In quantum field theory, it is well known that local Lagrangian dynamics can be consistent at the local level while harboring global anomalies — inconsistencies that become visible only when the theory is examined on topologically non-trivial backgrounds. The cancellation of global anomalies in the Standard Model (e.g., the requirement that the sum of hypercharges vanishes) is an example of a global consistency constraint that cannot be derived from the local Lagrangian alone. The SOA plays an analogous role in the Theory of Entropicity (ToE): it extends the variational framework beyond pointwise dynamics to ensure the coherence of the entropic field across the entire manifold.

Specifically, the SOA ensures that the spectrum of fluctuations of the entropy field — the eigenvalues and eigenfunctions of the entropic Laplacian on the manifold — satisfies global regularity, integrability, and boundary conditions. This spectral perspective connects the Theory of Entropicity (ToE) to the rich mathematical tradition of spectral geometry, where the geometry of a manifold is encoded in the spectrum of its Laplacian (the celebrated question, "Can one hear the shape of a drum?").

2.3 The Duality and Complementarity of LOA and SOA

Together, the LOA and SOA form a dual variational architecture that is, to the best of this author's knowledge, unique in the landscape of theoretical physics. The LOA handles the "local physics" — field propagation, matter coupling, local curvature generation — while the SOA handles the "global

physics" — spectral consistency, non-local correlations, and topological constraints. This duality mirrors, and arguably deepens, several familiar local/global dualities in established theories:

- In gauge theories, local gauge invariance (governed by the local Lagrangian) must be supplemented by global anomaly cancellation constraints (which are non-local and topological in character).
- In General Relativity, the local equivalence principle (local flatness of spacetime) coexists with global topological features (e.g., the distinction between simply connected and multiply connected spacetimes) that are not determined by local dynamics.
- In string theory, the local worldsheet dynamics (governed by the Polyakov action) are supplemented by global modular invariance constraints on the worldsheet partition function.

The LOA/SOA duality in the Theory of Entropicity (ToE) provides a single, unified framework for incorporating both local dynamics and global consistency — a feature that, if fully developed, would represent a significant structural advance.

2.4 The Self-Referential Character of the Obidi Action

Perhaps the most distinctive — and mathematically challenging — feature of the Obidi Action is its self-referential character. The metric $g_{ab}(S)$ depends on the field S , and the field S evolves according to equations derived from an action that depends on $g_{ab}(S)$. This creates a bootstrap structure: the geometry is not given in advance but is dynamically co-determined with the field that generates it.

This self-referential structure means that the Obidi Field Equations are inherently nonlinear and self-consistent: there is no fixed background

geometry against which the field evolves. The manifold's structure — its curvature, its topology, its causal structure — evolves dynamically with the entropic field. This background-free formulation is a defining feature of the Theory of Entropicity (ToE), and it sets the theory apart from both perturbative quantum field theory (which requires a fixed background metric) and classical General Relativity (which, while dynamical in its geometry, still presupposes a differentiable manifold as a starting point).

A useful comparison may be drawn with the Wheeler-DeWitt equation in canonical quantum gravity, which also possesses a background-free character — the wave functional $\Psi[h_{ij}]$ depends on the 3-geometry, and there is no external time parameter. However, the Wheeler-DeWitt equation achieves background independence through quantum geometry (quantization of the 3-metric), while the Theory of Entropicity (ToE) achieves it through entropic dynamics — a conceptually distinct route to a structurally similar destination.

Table 1: Comparison of Action Principles in Fundamental Physics

Theory	Action Principle	Fundament al Field	Backgroun d Dependenc e	Key Output
Newtonian Mechanics	Euler-Lagrange	Particle positions $q_i(t)$	Fixed (absolute space and time)	Newton's laws of motion
General Relativity	Einstein-Hilbert	Metric tensor $g_{\mu\nu}$	Background-free (dynamical geometry)	Einstein field equations
Quantum Field Theory (Standard Model)	Standard Model Lagrangian	Gauge and matter fields (A_μ, ψ, ϕ)	Fixed (Minkowski or curved background)	Feynman rules, scattering amplitudes

Theory	Action Principle	Fundamental Field	Background Dependence	Key Output
String Theory	Polyakov / Nambu-Goto	Worldsheet embedding $X^{\mu}(\sigma, \tau)$	Fixed target space (perturbative)	String spectrum, graviton
Loop Quantum Gravity	Holst / Palatini	Connection A_a , densitized triad E^a	Background-free	Spin networks, discrete spectra
Theory of Entropicity (ToE)	Obidi Action (LOA + SOA)	Entropy field $S(x)$	Background-free (self-referential)	Obidi Field Equations, emergent geometry

3. The Information-Geometric Foundations

The Theory of Entropicity (ToE) is not merely a theory that *uses* entropy; it is a theory that *geometrizes* entropy. The entropic field $S(x)$ does not propagate on an arbitrary mathematical manifold — it propagates on an information-geometric manifold, a space whose intrinsic geometry is determined by the structure of probability distributions, quantum states, and their interrelationships. This geometric substrate is not an optional decoration appended to the theory; it is the structural backbone that allows the Theory of Entropicity (ToE) to make rigorous contact with both classical and quantum physics from a single entropic foundation.

Three mathematical frameworks from information geometry and differential geometry converge within the Theory of Entropicity (ToE) to provide this substrate: the Fisher-Rao metric, the Fubini-Study metric, and the Amari-Čencov alpha-connection formalism.

3.1 The Fisher-Rao Metric

The Fisher-Rao metric, introduced by C.R. Rao in 1945 and independently characterized by N.N. Čencov, is the unique (up to a constant) Riemannian metric on the space of probability distributions that is invariant under sufficient statistics. For a parametric family of probability distributions $p(x; \theta)$, the Fisher-Rao metric is defined by the Fisher information matrix:

$$g_{ij}(\theta) = E[(\partial_i \ln p)(\partial_j \ln p)]$$

This metric encodes the distinguishability of nearby probability distributions: the larger the Fisher information, the more easily two neighboring distributions can be distinguished by observation. It provides a natural notion of "distance" in the space of probability distributions — the geodesic distance between two distributions reflects the minimum number of observations required to distinguish them.

In the Theory of Entropicity (ToE), the Fisher-Rao metric provides the geometric structure for the *classical sector* of the entropic manifold. Wherever the entropy field can be characterized by classical probability distributions — in the macroscopic, thermodynamic, or classical-statistical regime — the Fisher-Rao metric governs the geometry of the corresponding sector of the entropic manifold. This is directly relevant to the foundational claim of the Theory of Entropicity (ToE) that distinguishability is entropic: the Fisher-Rao metric quantifies precisely how much information (entropy) is required to distinguish nearby states.

3.2 The Fubini-Study Metric

The Fubini-Study metric is the natural Riemannian metric on the complex projective space CP^n — the space of pure quantum states (rays in Hilbert space). For two nearby quantum states $|\psi\rangle$ and $|\psi + d\psi\rangle$, the Fubini-Study distance is given by:

$$ds^2_{\text{FS}} = 1 - |\langle\psi|\psi + d\psi\rangle|^2$$

This metric encodes the distinguishability of nearby quantum states: two states that are nearly parallel in Hilbert space (high overlap) are close in the Fubini–Study metric; two states that are nearly orthogonal (low overlap) are far apart. The Fubini–Study metric is the quantum analog of the Fisher–Rao metric — indeed, it can be derived as the Fisher information metric for the probability distributions induced by quantum measurements.

In the Theory of Entropicity (ToE), the Fubini–Study metric provides the geometric structure for the *quantum sector* of the entropic manifold. Wherever the entropy field involves quantum superpositions, entanglement, or coherence — in the microscopic, quantum regime — the Fubini–Study metric governs the geometry of the corresponding sector. This dual classical-quantum geometric structure is a distinctive feature of the Theory of Entropicity (ToE): the entropic manifold is not uniformly classical or uniformly quantum, but possesses sectors whose geometry is governed by the appropriate information-geometric metric.

3.3 The Amari–Čencov Alpha-Connection Formalism

The alpha-connection, developed by Shun-ichi Amari and rooted in Čencov's invariance theorems, provides a one-parameter family of affine connections $\nabla^{(\alpha)}$ on statistical manifolds. For each value of the parameter $\alpha \in \mathbb{R}$, one obtains a distinct connection — and hence a distinct notion of parallel transport, geodesic, and curvature — on the space of probability distributions.

Three values of α are of particular significance:

- **$\alpha = 0$:** The Levi-Civita connection — the unique torsion-free, metric-compatible connection of Riemannian geometry. At this value, the information-geometric manifold reduces to a standard Riemannian manifold, and the standard tools of differential geometry apply.

- **$\alpha = +1$:** The exponential connection (e-connection) — naturally associated with exponential families of probability distributions and maximum likelihood estimation.
- **$\alpha = -1$:** The mixture connection (m-connection) — naturally associated with mixture families and Bayesian inference.

In the Theory of Entropicity (ToE), the Amari-Čencov formalism provides the crucial bridge between the classical (Fisher-Rao) and quantum (Fubini-Study) geometries. The entropic order parameter α emerges within the theory as a universal deformation index linking geometry, information, and entropy flow. As α varies continuously, the geometry of the entropic manifold interpolates between different informational regimes — from the flat, Euclidean geometry of maximum-entropy (equilibrium) states to the highly curved geometry of far-from-equilibrium, highly structured configurations.

Furthermore, the Theory of Entropicity (ToE) incorporates generalized entropy measures — specifically Rényi entropy $H_\alpha = (1/(1-\alpha)) \ln \sum p_i^\alpha$ and Tsallis entropy $S_q = (1/(q-1))(1 - \sum p_i^q)$ — and establishes a correspondence between these generalized entropies and the alpha-connection geometry. This is a non-trivial mathematical achievement: it embeds the landscape of generalized entropies (which have found applications in complex systems, non-extensive statistical mechanics, and quantum information) within the unified geometric framework of the entropic manifold.

3.4 Synthesis

By integrating the Fisher-Rao metric, the Fubini-Study metric, and the Amari-Čencov alpha-connection through the Obidi Action, the Theory of Entropicity (ToE) provides a rigorous geometric and probabilistic structure for physical evolution within an entropic manifold. The classical and quantum sectors of the manifold are not treated as separate domains requiring separate theories (as in the current fragmentation of physics into General

Relativity and quantum field theory), but as different geometric regimes of a single information-geometric substrate — related by the continuous deformation parameter α .

This is not merely decorative mathematics. It is the structural backbone that allows the Theory of Entropicity (ToE) to claim, at the level of mathematical architecture, a unification of classical and quantum physics within a single variational framework. Whether this claim survives the test of explicit calculation is a question to which we shall return.

4. The Obidi Field Equations and the Master Entropic Equation

4.1 Derivation from the Obidi Action

The Obidi Field Equations (OFE) arise from the standard variational procedure applied to the Obidi Action: the requirement that the action be stationary under arbitrary variations of the entropy field, $\delta A[S] = 0$. This is the same procedure that yields the Euler-Lagrange equations in classical mechanics, the Einstein field equations from the Einstein-Hilbert action, and the Yang-Mills equations from the gauge field action. In this sense, the derivation of the OFE follows a well-trodden mathematical path.

However, the resulting equations are significantly more complex than their analogs in other theories, precisely because of the self-referential character of the Obidi Action. In General Relativity, the Einstein-Hilbert action $S_{\text{EH}} = \int R \sqrt{|g|} d^4x$ depends on the metric $g_{\mu\nu}$, and variation with respect to $g_{\mu\nu}$ yields the Einstein equations. The key point is that the metric is the fundamental variable: one varies the action with respect to $g_{\mu\nu}$ and obtains equations for $g_{\mu\nu}$. In the Theory of Entropicity (ToE), the situation is more intricate. The fundamental variable is the entropy field $S(x)$, but the metric $g_{ab}(S)$ also

depends on S . Variation of the Obidi Action with respect to S therefore produces terms involving both S and its first and second derivatives, as well as terms involving the variation of the metric with respect to S — yielding a coupled, nonlinear system of equations.

4.2 Structure and Character

The Obidi Field Equations possess three essential characteristics that distinguish them from the dynamical equations of other fundamental theories:

(a) Dynamic: Each update of the entropy field $S(x)$ modifies the induced geometry $g_{ab}(S)$, which in turn modifies the equations governing the evolution of S . The equations and their solutions co-evolve — a feature sometimes called "co-evolution of law and state" in the philosophy of physics. This dynamic character is more radical than the dynamical geometry of General Relativity, where the metric evolves but the form of the Einstein equations remains fixed. In the Theory of Entropicity (ToE), the effective form of the equations changes as the geometry changes, because the coefficients in the equations depend on the geometry, which depends on the field.

(b) Probabilistic: The entropy field $S(x)$ lives on an information-geometric manifold whose structure reflects the geometry of probability distributions. The Fisher-Rao and Fubini-Study metrics endow the field space with a probabilistic geometry. Consequently, the OFE are not merely differential equations for a scalar field; they are differential equations on a statistically structured manifold, encoding the distinguishability, correlation, and information content of the field configuration at every point.

(c) Algorithmic: The self-referential, iterative character of the OFE invites an interpretation of the evolution of $S(x)$ as a continuous, self-correcting computation. The field "computes" its own geometry, which "computes" the field equations, which "compute" the next state of the field. This algorithmic

interpretation resonates with the recent trend in theoretical physics toward computational and information-theoretic formulations of fundamental physics — exemplified by the work of Seth Lloyd, Leonard Susskind, and Juan Maldacena on computational complexity and the structure of spacetime.

Because of their self-referential character, the OFE do not admit closed-form analytical solutions in any physically meaningful regime. Solutions must be obtained through iterative, adaptive, and computational procedures. This is not, in itself, a fatal limitation — the Einstein field equations of General Relativity are also nonlinear and generally do not admit closed-form solutions except in highly symmetric cases (Schwarzschild, Kerr, FLRW). The majority of physically relevant solutions in General Relativity are obtained numerically. The same is expected for the OFE.

4.3 Relationship to the Master Entropic Equation (MEE)

The Obidi Field Equations and the Master Entropic Equation (MEE) are formally equivalent designations for the same system of equations. The term "Master Entropic Equation" emphasizes the equation's role as the single, overarching dynamical equation from which all physical evolution — classical, quantum, gravitational, thermodynamic — is claimed to emerge. The term "Obidi Field Equations" emphasizes the variational origin and the field-theoretic character of the equations.

A feature of the MEE that merits particular attention is its incorporation of irreversibility directly into the structure of physical law. The conventional wave equations of physics — the Schrödinger equation, the Klein-Gordon equation, the Dirac equation, Maxwell's equations — are all time-reversal symmetric (or CPT-symmetric). They do not, at the level of fundamental dynamics, distinguish between past and future. The arrow of time in conventional physics must be imposed externally, typically through boundary

conditions or statistical arguments (the Boltzmann H -theorem, the past hypothesis). The MEE, by contrast, incorporates irreversibility at the fundamental level through the entropic structure of the manifold and the directional character of the Amari-Čencov connections. The arrow of time is not imposed from outside; it emerges from the asymmetric geometry of the information-geometric manifold on which the entropy field evolves.

This resolution of the time-reversal paradox at the level of fundamental dynamics — if sustained under rigorous mathematical scrutiny — would constitute one of the most significant conceptual advances in the Theory of Entropicity (ToE).

4.4 The Iterative Solution Architecture

The self-referential character of the OFE necessitates an iterative approach to obtaining solutions. The Theory of Entropicity (ToE) envisions the following schematic solution procedure: (1) Begin with an initial configuration of the entropy field $S_0(x)$. (2) Compute the induced metric $g_{ab}(S_0)$. (3) Solve the OFE on the manifold equipped with this metric to obtain an updated field $S_1(x)$. (4) Recompute the metric $g_{ab}(S_1)$. (5) Iterate until convergence is achieved — until $S_{n+1} \approx S_n$ to within a specified tolerance.

This iterative structure raises important mathematical questions concerning convergence, uniqueness, and stability of solutions. Under what conditions does the iterative procedure converge? Is the fixed point unique, or can different initial configurations converge to different solutions? How sensitive is the final solution to the choice of initial configuration? These are questions of both mathematical interest and physical significance — they bear directly on the predictive power of the theory. The Theory of Entropicity (ToE) has not yet provided rigorous answers to these questions, and they represent an important open frontier for the research program.

5. The Obidi Curvature Invariant (OCI) — The Quantum of Distinguishable Curvature

5.1 Definition and Origin

Among the most striking predictions of the Theory of Entropicity (ToE) is the existence of a fundamental curvature invariant: the Obidi Curvature Invariant (OCI), defined as:

$$\text{OCI} = \ln 2$$

The OCI arises not from an arbitrary definition or a parameter fit, but from the internal mathematical structure of the theory. Specifically, it emerges as the first non-zero minimum of a distinguishability potential defined on the informational-entropic manifold. This potential quantifies the "entropic cost" of distinguishing two nearby states on the manifold; the OCI represents the smallest entropic curvature gap at which two states become meaningfully — physically, operationally — distinct.

The mathematical derivation proceeds from the information-geometric structure of the entropic manifold: the Fisher-Rao metric defines a notion of distance between probability distributions, and the curvature of this metric quantifies how rapidly distinguishability changes as one moves through the space of states. The OCI is the minimal curvature at which a binary distinction — the simplest possible distinction, between two states — first becomes possible.

5.2 Mathematical Character

The value $\ln 2$ is not a coordinate-dependent artifact. Because it arises as a minimum of a coordinate-independent potential defined on the information-

geometric manifold, the OCI behaves as a true scalar invariant in differential geometry. It does not depend on the choice of coordinates, parametrizations, or representations — it is a property of the manifold itself, not of any particular description of the manifold.

In this sense, the OCI is analogous to the canonical scalar invariants of classical differential geometry — such as the Kretschner scalar $K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$ or the Gauss-Bonnet invariant — which characterize the curvature of a manifold in a coordinate-independent manner. However, while the Kretschner scalar characterizes the geometry of spacetime, the OCI characterizes the geometry of the entropic field — it is a curvature invariant of the *information-geometric* manifold, not of the spacetime manifold per se.

5.3 Physical Interpretation

The physical interpretation of the OCI is both elegant and profound. The OCI = $\ln 2$ represents the smallest entropic curvature needed for one physical state to be meaningfully different from another. It is, in the language of the Theory of Entropicity (ToE), the fundamental quantum of distinguishable curvature — the minimal geometric structure required for a binary distinction to exist.

The corresponding binary distinguishability relation takes a simple and striking form:

$$\rho_B = 2\rho_A$$

where ρ_A and ρ_B are two entropic density states, and the factor of 2 represents the minimal separation between two distinguishable states. This factor of 2 is not arbitrary — it is the information-theoretic consequence of the binary nature of the simplest possible distinction (one bit of information corresponds to $\ln 2$ nats of entropy).

The OCI thus serves as a bridge between three domains: information theory (where $\ln 2$ is the entropy of a binary choice), geometry (where it appears as a

curvature invariant), and physics (where it marks the onset of discernible physical change). This triple connection — information, geometry, physics — is a hallmark of the Theory of Entropicity (ToE).

5.4 Significance

The significance of the OCI extends in several directions:

Natural ultraviolet cutoff: If the OCI represents the minimum meaningful curvature in the entropic description of reality, it may provide a natural ultraviolet cutoff for quantum field theories — resolving, or at least ameliorating, the divergence problems that plague perturbative quantum field theory. Curvature fluctuations below the OCI scale would be physically meaningless (no distinction can be drawn), effectively imposing a minimum length scale below which field-theoretic descriptions lose their applicability.

Connection to the Landauer limit: The Landauer limit in information thermodynamics establishes that the erasure of one bit of information requires a minimum energy dissipation of $k_B T \ln 2$. The appearance of $\ln 2$ in both the OCI and the Landauer limit is suggestive of a deep connection between computational irreversibility and entropic geometry — a connection that the Theory of Entropicity (ToE) is uniquely positioned to explore.

Uniqueness: It bears emphasis that the OCI is a uniquely ToE prediction. No other framework in current theoretical physics — neither General Relativity, nor quantum field theory, nor string theory, nor loop quantum gravity — produces this invariant from first principles. If experimentally confirmed or if its consequences can be independently verified, the OCI would constitute a distinctive empirical signature of the Theory of Entropicity (ToE).

6. The Vuli-Ndlela Integral — Entropy-Constrained Quantum Mechanics

6.1 Conceptual Foundation

The Vuli-Ndlela Integral is an entropy-constrained and entropy-weighted reformulation of Feynman's path integral — one of the most elegant and powerful formulations of quantum mechanics. In Feynman's original formulation, the quantum amplitude for a process is computed by summing over *all* possible paths (histories) connecting the initial and final states, with each path weighted by the exponential of the classical action:

$$\langle x_f | x_i \rangle = \int D[x(t)] \exp(iS_{cl}[x]/\hbar)$$

In this formulation, all paths contribute — including wildly non-classical paths that deviate enormously from the classical trajectory. The interference among all these paths produces the quantum amplitude, and the classical limit emerges through the stationary-phase approximation (the paths near the classical trajectory dominate when $\hbar \rightarrow 0$).

The Vuli-Ndlela Integral modifies this formulation in a fundamental way: paths are not merely weighted by the classical action, but are additionally weighted and *constrained* by entropic considerations. Specifically, paths that incur large entropy costs — paths that would gratuitously generate entropy, violate entropic bounds, or transgress the Vuli-Ndlela inequality — are exponentially suppressed. Nature, in the Theory of Entropicity (ToE), favors trajectories that are entropically economical.

6.2 Mathematical Structure

The Vuli-Ndlela Integral incorporates three key elements beyond the standard Feynman path integral:

- **The classical action $S_{\text{cl}}[\varphi]$** — retained from the standard formulation, providing the phase factor that governs quantum interference.
- **The gravitational entropy $S_{\text{G}}[\varphi]$** — an entropy contribution associated with the gravitational (geometric) content of the path. This term ensures that the path integral respects the entropic constraints imposed by gravity — connecting to Bekenstein's entropy bound and the holographic principle.
- **The irreversibility entropy $S_{\text{irr}}[\varphi]$** — an entropy contribution associated with the irreversibility of the path. This is the term that breaks time-reversal symmetry at the level of the path integral: paths that proceed in the thermodynamically favored direction (increasing total entropy) are enhanced, while paths that proceed against the thermodynamic arrow are suppressed.

Integration is performed not over all conceivable histories (as in the unrestricted Feynman integral) but over a restricted set of histories where entropy densities and fluxes obey the Vuli-Ndlela inequality. This inequality provides an upper bound on the entropic cost of any physically realizable path — effectively excluding from the path integral those histories that would violate the entropic structure of the theory.

6.3 Physical Consequences

The Vuli-Ndlela Integral has several far-reaching physical consequences that distinguish it from the standard Feynman path integral:

The quantum measurement problem: In the standard formulation of quantum mechanics, wave function collapse — the transition from a superposition of states to a definite outcome upon measurement — is postulated, not derived. It is an axiom of the theory, not a consequence of the dynamics. The Vuli-Ndlela Integral offers a resolution: wave function collapse emerges as an entropy-driven phase transition. When a quantum system interacts with a macroscopic measuring apparatus, the entropic constraints imposed by the Vuli-Ndlela inequality rapidly suppress the superposition of macroscopically distinct states — not because of a separate "collapse postulate," but because superpositions of macroscopically distinct states are entropically unsustainable.

Irreversibility and temporal asymmetry: The inclusion of $S_{\text{irr}}[\varphi]$ introduces irreversibility directly into the quantum formalism. The path integral is no longer time-reversal symmetric — it distinguishes between forward-in-time and backward-in-time evolution. This provides a natural origin for the quantum arrow of time, grounded in the entropic structure of the theory rather than in external boundary conditions.

Decoherence and the classical-quantum transition: The entropic weighting provides a natural mechanism for decoherence — the progressive loss of quantum coherence as a system interacts with its environment. In the Vuli-Ndlela framework, decoherence is not merely an environmental effect (as in the standard decoherence program of Zurek and others) but a consequence of the fundamental entropic structure of quantum evolution.

6.4 Relationship to Standard Path Integrals

A critical requirement for any extension of quantum mechanics is that it reduce to the standard formulation in the appropriate limit. The Theory of Entropicity (ToE) claims that the Vuli-Ndlela Integral satisfies this requirement: in the limit where entropic constraints are relaxed — where $S_G \rightarrow 0$, $S_{\text{irr}} \rightarrow 0$, and the Vuli-Ndlela inequality becomes non-binding — the Vuli-

Ndlela Integral reduces to the standard Feynman path integral. This ensures backward compatibility with standard quantum mechanics while extending it with entropic structure. The mathematical demonstration of this limit, in full rigor, remains an important open task for the theory.

7. Reproduction of Einstein's Field Equations

7.1 The Entropic Derivation of Gravity

Any theory that claims to subsume General Relativity must demonstrate, with mathematical rigor, that Einstein's field equations emerge as a limiting case of its own dynamical equations. The Theory of Entropicity (ToE) makes precisely this claim: the Einstein field equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$$

are recovered as a limiting case of the Obidi Field Equations in the regime where the entropy field is slowly varying, the entropic curvature is small, and the self-referential corrections are negligible. In this regime, the entropic metric $g_{ab}(S)$ reduces to a standard Lorentzian metric, the entropic curvature scalar $R(S)$ reduces to the Ricci scalar R , and the Obidi Action reduces to the Einstein-Hilbert action.

The conceptual picture is as follows: gravity, in the Theory of Entropicity (ToE), is not a fundamental force. It is an emergent phenomenon arising from entropy-driven constraints on the configurations of matter and energy. Objects move along paths that maximize entropy flow — trajectories that are, in the appropriate limit, geodesics of the emergent spacetime geometry. Spacetime curvature is the geometric manifestation of local entropic gradients: where entropy varies rapidly, the emergent geometry is highly curved; where entropy is uniform, the geometry is flat. Mass, in this picture,

is the visible residue of entropy concentration — a region of high entropy density generates a region of high curvature, which we interpret as the gravitational field of a massive body.

This picture is not entirely novel — it has antecedents in the work of Jacobson (1995), who derived the Einstein equations from the Clausius relation applied to local causal horizons; Verlinde (2011), who proposed that gravity is an entropic force; and Bianconi (2025), who derived modified Einstein equations from quantum relative entropy. The Theory of Entropicity (ToE) extends and deepens these approaches by deriving the spacetime geometry itself from the entropic field, rather than presupposing it.

7.2 Specific Classical Tests

The Theory of Entropicity (ToE) claims to reproduce the classical tests of General Relativity through entropic dynamics. Two such tests are particularly significant:

The perihelion precession of Mercury: The anomalous advance of Mercury's perihelion — 43 arcseconds per century, unexplained by Newtonian gravity — was one of the founding empirical successes of General Relativity. The Theory of Entropicity (ToE) claims to reproduce this result through the entropic framework, with the entropic coupling constant η playing a role analogous to Newton's gravitational constant G . The entropic curvature generated by the Sun's mass distribution produces a correction to the Keplerian orbit that, in the appropriate limit, yields the same 43-arcsecond precession.

The deflection of starlight: The bending of light by the Sun — measured at 1.75 arcseconds during the 1919 Eddington expedition — is similarly derivable within the entropic framework. The entropic gradient generated by the Sun acts on photon trajectories in a manner that reproduces the General Relativistic prediction.

These are critical tests: any theory claiming to subsume General Relativity must reproduce its verified predictions quantitatively, not merely qualitatively. The claim of exact quantitative reproduction requires detailed, step-by-step mathematical derivation, which has been presented in preliminary form and awaits full peer-reviewed scrutiny.

7.3 The Modified Einstein Field Equations

Beyond the standard Einstein equations, the Theory of Entropicity (ToE) introduces modifications that become significant in extreme regimes. The modified equations take the schematic form:

$$G_{\mu\nu} + \Lambda_{\text{eff}}(S) g_{\mu\nu} + J_{\mu\nu}(S) = (8\pi G/c^4) T_{\mu\nu}$$

where $\Lambda_{\text{eff}}(S)$ is an effective cosmological term that replaces the constant Λ with a dynamically evolving entropy potential $\mathcal{V}(\Lambda)$, and $J_{\mu\nu}(S)$ is a kinetic entropy flux term capturing the flow of entropy across the manifold. These modifications are small in the weak-field, low-curvature regime — ensuring compatibility with the extensive experimental verification of General Relativity in the solar system — but become significant in extreme regimes: near black hole horizons, in the early universe, and at cosmological scales. It is in these extreme regimes that the Theory of Entropicity (ToE) makes predictions that differ from standard General Relativity and standard cosmology.

8. The Generalized Entropic Expansion Equation (GEEE) and Entropic Cosmology

8.1 The GEEE

The Generalized Entropic Expansion Equation (GEEE) is an entropy-driven modification to the Friedmann equations of standard cosmology. The Friedmann equations — derived from the Einstein field equations applied to a homogeneous, isotropic universe (the FLRW metric) — are the cornerstone of modern cosmology. They relate the expansion rate of the universe (the Hubble parameter H) to the energy density ρ , pressure p , and cosmological constant Λ :

$$H^2 = (8\pi G/3)\rho - k/a^2 + \Lambda/3$$

The GEEE modifies these equations by replacing the cosmological constant Λ with a dynamically evolving entropy-driven term $\Lambda_{\text{ent}}(S, a)$ that emerges from the entropic field's evolution. The resulting equation describes both the deceleration and acceleration of the universe's expansion as natural consequences of entropic evolution — without invoking any exotic "dark energy" component.

The GEEE thus provides a unified entropic account of the entire expansion history of the universe: the early deceleration (dominated by matter and radiation, with the entropic term subdominant) and the late-time acceleration (dominated by the entropic term, which grows as the universe expands and entropy increases).

8.2 Resolution of the Dark Energy Problem

The dark energy problem is arguably the most profound crisis in modern cosmology. Observations of Type Ia supernovae, the cosmic microwave background, and baryon acoustic oscillations all indicate that the expansion of the universe is accelerating. In the standard Λ CDM cosmological model, this acceleration is attributed to a cosmological constant Λ — interpreted as the vacuum energy density of quantum fields. However, quantum field theory predicts a vacuum energy density that exceeds the observed value by approximately 120 orders of magnitude — a discrepancy that has been called "the worst theoretical prediction in the history of physics."

The GEEE offers a fundamentally different resolution. In the Theory of Entropicity (ToE), cosmic acceleration is not driven by a mysterious vacuum energy; it is a natural thermodynamic consequence of entropic evolution. As the universe expands, the entropy of the universe increases (in accordance with the second law of thermodynamics), and the entropic term $\Lambda_{\text{ent}}(S, a)$ in the GEEE grows in a manner that produces late-time acceleration. The cosmological constant is not a fundamental parameter that must be explained (and whose quantum field theoretic prediction fails by 120 orders of magnitude) — it is an effective, emergent quantity whose value is determined by the entropic state of the universe.

If the GEEE can reproduce the observed expansion history of the universe — the deceleration at high redshift, the transition to acceleration at $z \sim 0.7$, and the observed value of the effective cosmological constant — with the same quantitative precision as Λ CDM, it would represent a major triumph of the entropic approach, eliminating one of the most troubling fine-tuning problems in modern physics.

8.3 The Entropic Acceleration Threshold (EAT)

The GEEE introduces a novel concept: the Entropic Acceleration Threshold (EAT). This is a critical entropy density above which the universe transitions from decelerated expansion (dominated by gravitational attraction) to accelerated expansion (dominated by entropic effects). The EAT provides a natural, physical mechanism for the observed transition epoch — the moment in cosmic history when the universe's expansion shifted from decelerating to accelerating, observed to occur at approximately $z \sim 0.7$ (about 5 billion years ago).

In standard cosmology, this transition is explained by the crossing point at which the constant dark energy density begins to dominate over the declining matter density. In the GEEE, the transition is explained by the crossing of the Entropic Acceleration Threshold — the entropy density of the universe reaches a critical value at which entropic effects begin to dominate the expansion dynamics. The EAT is not a free parameter to be fitted to observations; it emerges from the structure of the GEEE and the entropic evolution of the universe.

8.4 Dark Matter Reinterpretation

The Theory of Entropicity (ToE) also offers a reinterpretation of dark matter phenomenology. In standard cosmology, the observed rotation curves of galaxies, the dynamics of galaxy clusters, gravitational lensing, and the cosmic microwave background all point to the existence of a non-luminous, non-baryonic matter component — "dark matter" — comprising approximately 27% of the universe's energy budget. Despite decades of experimental effort, no dark matter particle has been directly detected.

In the entropic framework, the effects attributed to dark matter may be attributable — at least in part — to entropic curvature and gradients within

the entropic field. Regions of enhanced entropy density produce enhanced entropic curvature, which mimics the gravitational effects of unseen matter. This approach shares certain features with Modified Newtonian Dynamics (MOND) and with Verlinde's emergent gravity, but is grounded in the more comprehensive framework of the Obidi Field Equations and the GEEE. Whether this reinterpretation can quantitatively account for the full range of dark matter phenomenology remains an open and critical question.

9. Subsumption of Bianconi's "Gravity from Entropy" (GfE)

9.1 Bianconi's GfE Framework

In 2025, Professor Ginestra Bianconi of Queen Mary University of London published "Gravity from Entropy" in *Physical Review D* (volume 111, 066001), proposing an entropic action that couples matter fields with geometry through quantum relative entropy. Bianconi's framework is mathematically rigorous, physically motivated, and published in one of the most prestigious journals in theoretical physics. It merits serious attention from any review of entropic approaches to gravity.

The key features of Bianconi's GfE framework are as follows. The spacetime metric $g_{\mu\nu}$ is treated not as a classical field but as a quantum operator — specifically, as an effective density matrix describing the quantum state of the geometry. Modified Einstein equations are derived from the minimization of quantum relative entropy between the actual quantum state of the geometry and a reference state. A Lagrangian multiplier field — the G-field — emerges from the variational procedure and produces an emergent, small, positive cosmological constant. The framework addresses the cosmological constant problem from an information-theoretic perspective and makes contact with both semiclassical gravity and quantum information theory.

9.2 How the Theory of Entropicity (ToE) Subsumes GfE

The Theory of Entropicity (ToE) claims — and this Letter examines the argument critically — that Bianconi's GfE framework is subsumed as a special instance within the broader architecture of the Theory of Entropicity (ToE). The subsumption argument proceeds as follows:

The Spectral Obidi Action (SOA), which governs the global, spectral, and non-local consistency of the entropic field, contains — as a special limiting case — a structure that is mathematically equivalent to Bianconi's entropic action. Specifically, in the limit where: (i) the entropic manifold is restricted to a fixed Lorentzian background (breaking the background-free character of full ToE); (ii) the entropy field $S(x)$ is identified with the quantum relative entropy between the geometric state and a reference state; and (iii) the self-referential coupling between field and geometry is linearized — the SOA reduces to Bianconi's entropic action.

In this view, GfE is the "fixed-background specialization" of a more general background-free entropic framework. Where Bianconi's framework starts from a pre-existing Lorentzian spacetime and introduces an entropic action on top of it, the Theory of Entropicity (ToE) goes further by deriving the spacetime geometry itself from the entropic field. GfE, in this reading, is the Theory of Entropicity (ToE) with the self-referential coupling turned off and the background geometry frozen.

9.3 Critical Evaluation

This is a strong claim that requires careful and dispassionate scrutiny. Let us assess both the structural parallels and the critical differences.

The structural parallels are genuine and substantial. Both frameworks derive gravity from entropic principles. Both produce modified Einstein equations. Both address the cosmological constant problem. Both employ information-

theoretic tools (relative entropy, Fisher information, density matrices) as central mathematical objects. The conceptual alignment between GfE and the Theory of Entropicity (ToE) is evident and goes beyond superficial similarity. However, there are critical differences that bear on the subsumption claim. Bianconi's framework is fully mathematically explicit: every step of the derivation is presented in complete detail, the variational procedure is carried out rigorously, and the resulting modified Einstein equations are written in closed form. The Theory of Entropicity (ToE), by contrast, operates at a higher level of generality but with correspondingly less explicit mathematical detail. The Obidi Action is defined schematically, and the precise functional forms of its components are still under development. The explicit mathematical reduction — the step-by-step demonstration that Bianconi's action emerges as a limit of the SOA — has not yet been carried out in full detail.

Furthermore, Bianconi's framework has undergone peer review in a top-tier journal, while the Theory of Entropicity (ToE) has been presented primarily through preprints and open-access platforms. This does not invalidate the Theory of Entropicity (ToE), but it does mean that the subsumption claim currently rests more on conceptual-structural arguments than on rigorously demonstrated mathematical reduction.

For full acceptance of the subsumption claim, the Theory of Entropicity (ToE) would need to exhibit an explicit mathematical reduction: starting from the SOA with fully specified functional forms, imposing the limiting conditions described above, and arriving — step by step — at Bianconi's entropic action. Until this is accomplished, the subsumption claim remains a plausible conjecture rather than a demonstrated theorem.

Table 2: Comparison of the Theory of Entropicity (ToE) with Bianconi's Gravity from Entropy (GfE)

Feature	Theory of Entropicity (ToE)	Gravity from Entropy (GfE)
Ontological status of entropy	Fundamental dynamical field — the sole ontological primitive from which all physics emerges	Central information-theoretic tool — quantum relative entropy drives geometry, but spacetime is presupposed
Starting point	Entropic field $S(x)$ on an information-geometric manifold; no pre-existing spacetime	Pre-existing Lorentzian spacetime manifold with quantum fields
Spacetime treatment	Emergent — derived from the entropic field; background-free and self-referential	Presupposed — Lorentzian manifold given; metric treated as quantum operator (density matrix)
Action principle	Obidi Action (LOA + SOA) — dual variational structure with local and spectral components	Entropic action — single variational functional based on quantum relative entropy
Modified Einstein equations	Yes — with kinetic entropy flux and entropy potential corrections	Yes — with G-field Lagrangian multiplier corrections
Cosmological constant	Emergent and dynamically evolving through the GEEE; resolves the 120 orders-of-magnitude problem	Emergent and small; produced by the G-field mechanism
Dark matter	Reinterpreted as entropic curvature effects — no new particles required	Not directly addressed in the published framework
Peer review status	Preprints and open-access publications (Cambridge Open Engage, ResearchGate)	Peer-reviewed, published in <i>Physical Review D</i> (2025)
Mathematical	Conceptual	Fully explicit — all

Feature	Theory of Entropicity (ToE)	Gravity from Entropy (GfE)
explicitness	architecture fully articulated; explicit functional forms under development	derivations carried out in complete mathematical detail

10. Critical Assessment and Open Questions

A serious review must be critical as well as expository. It is the obligation of the reviewer — particularly in the assessment of a framework as ambitious as the Theory of Entropicity (ToE) — to distinguish between what has been achieved, what has been plausibly argued, and what remains to be demonstrated. The following assessment is offered in a spirit of constructive rigor.

10.1 Mathematical Formalization Gaps

The most significant challenge facing the Theory of Entropicity (ToE) at its current stage of development is the gap between conceptual architecture and explicit mathematical formalization. The conceptual architecture — the Obidi Action, the dual LOA/SOA structure, the information-geometric foundations, the OFE/MEE, the OCI, the Vuli-Ndlela Integral, the GEEE — is structurally coherent and conceptually powerful. But several critical components remain under development in their explicit mathematical detail:

- The explicit functional forms of $A(S)$, $V(S)$, and the coupling term $\eta S T^{\text{pv}}$ within the LOA are specified schematically but not in fully explicit, closed form. Without these functional forms, the Obidi Action cannot be varied to produce explicit Obidi Field Equations.

- The derivation of the OFE from the Obidi Action, while conceptually clear, has not been presented with the step-by-step mathematical rigor that the theoretical physics community requires for acceptance — the kind of rigor exhibited, for example, in the derivation of the Einstein equations from the Einstein-Hilbert action in any standard textbook on General Relativity.
- The exact forms of the gravitational entropy $S_G[\varphi]$ and the irreversibility entropy $S_{\text{irr}}[\varphi]$ in the Vuli-Ndlela Integral require explicit specification and mathematical analysis.
- The Spectral Obidi Action, while conceptually motivated, is the least mathematically developed component of the framework. Its precise mathematical content — the spectral conditions, the global constraints, the relationship to the LOA — requires substantial further elaboration.

These gaps are not fatal — they are characteristic of a theory in an early stage of development. General Relativity itself underwent years of mathematical development between Einstein's initial publication in 1915 and the rigorous mathematical formulations of the theory by Hilbert, Weyl, and others. But the gaps must be acknowledged and addressed.

10.2 Solution Space

The self-referential character of the OFE raises important questions about the solution space of the theory. What convergence guarantees exist for the iterative solution procedure? Is the fixed point unique, or can the theory admit multiple self-consistent solutions for the same physical situation (a many-solutions problem analogous to the landscape problem in string theory)? What are the stability properties of solutions — do small perturbations in the initial entropy field configuration lead to small perturbations in the solution, or can the nonlinear, self-referential dynamics

amplify initial perturbations? These are questions of mathematical analysis that have not yet been addressed within the framework.

10.3 Experimental Predictions

A physical theory is ultimately judged by its experimental predictions. The Theory of Entropicity (ToE) has pointed to the 232-attosecond constraint on entanglement formation — derived from experimental data in attosecond physics — as empirical evidence for the Entropic Time Limit (ETL). This is a promising start, but the Theory of Entropicity (ToE) needs to generate unique, quantitative predictions that distinguish it from the combined framework of standard quantum field theory and General Relativity. Predictions that are merely consistent with existing observations (which are already explained by established theories) are insufficient; what is needed are predictions of new phenomena, or predictions of known phenomena with different quantitative values, that can be tested experimentally.

Potential avenues for distinctive predictions include: modifications to gravitational wave signatures in extreme curvature regimes (testable by LIGO/Virgo/KAGRA and future detectors); modifications to the cosmic microwave background power spectrum (testable by current and future CMB experiments); distinctive signatures in black hole thermodynamics; and entropic corrections to quantum interference experiments.

10.4 Relationship to Established Frameworks

The precise mathematical pathway from the Obidi Action to the Einstein field equations — claimed but not yet demonstrated in full, explicit detail — is essential. Similarly, the derivation of the quantum uncertainty relations from the MEE requires rigorous mathematical demonstration. The Theory of Entropicity (ToE) claims to subsume both General Relativity and quantum mechanics; for these claims to be credible, the derivation of both frameworks

as limiting cases of the OFE must be carried out with the same mathematical rigor as the original derivations of these theories.

10.5 The Path Forward

The research program defined by the Theory of Entropicity (ToE) has a clear path forward, defined by the following priorities:

1. **Computational physics:** Numerical solutions of the OFE in simple, highly symmetric cases (spherically symmetric static solutions, cosmological solutions) would provide the first concrete demonstration that the equations can be solved and produce physically reasonable results.
2. **Explicit demonstration of limiting cases:** The derivation of the Einstein field equations and quantum uncertainty relations from the OFE, carried out with full mathematical rigor, is essential for the credibility of the subsumption claims.
3. **Novel experimental predictions:** The identification of unique, quantitative predictions that differ from standard QFT+GR — preferably predictions that are testable with current or near-future experimental capabilities.
4. **Mathematical rigor:** Full specification of all functional forms in the Obidi Action, rigorous derivation of the OFE, and analysis of the solution space (existence, uniqueness, stability).
5. **Peer-reviewed publication:** Submission of the core mathematical framework to peer-reviewed journals in mathematical physics and general relativity. Peer review is not merely a gatekeeping exercise — it is a process of critical scrutiny that strengthens and refines theoretical frameworks.

11. Concluding Remarks

This Letter has undertaken a systematic examination of the mathematical architecture of the Theory of Entropicity (ToE). We have reviewed the Obidi Action and its dual LOA/SOA structure — a variational architecture that is, to the best of this author's knowledge, unique in theoretical physics in its combination of local and spectral, differential and global components. We have examined the information-geometric foundations — the Fisher-Rao metric, the Fubini-Study metric, and the Amari-Čencov alpha-connection — that provide the rigorous geometric substrate for the entropic field. We have analyzed the Obidi Field Equations and the Master Entropic Equation, noting their dynamic, probabilistic, and algorithmic character and their incorporation of irreversibility at the fundamental level. We have assessed the Obidi Curvature Invariant ($\text{OCI} = \ln 2$) as a novel and distinctively ToE prediction — a fundamental quantum of distinguishable curvature with connections to information theory, geometry, and physics. We have reviewed the Vuli-Ndlela Integral as an entropy-constrained reformulation of the Feynman path integral that addresses the measurement problem, decoherence, and temporal asymmetry. We have examined the claimed reproduction of Einstein's field equations and the classical tests of General Relativity. We have analyzed the GEEE and its implications for entropic cosmology without dark energy. And we have critically evaluated the subsumption of Bianconi's Gravity from Entropy framework.

The assessment is, necessarily, mixed — as befits a theory in an active stage of development. The architecture is ambitious, structurally coherent, and conceptually powerful. The Theory of Entropicity (ToE) occupies a unique position in the landscape of theoretical physics: it is more radical than entropic gravity (Verlinde, Jacobson, Bianconi) because it derives geometry itself from entropy, yet it employs the familiar and proven mathematical language of variational principles, field equations, and path integrals. It seeks

not to replace the mathematical tools of theoretical physics but to rebuild them on a different ontological foundation — entropy rather than geometry.

The critical gaps in explicit mathematical formalization are acknowledged. These are not weaknesses that invalidate the framework, but challenges that define the research program's next phase. Every major theory in physics has passed through a stage of conceptual maturity preceding full mathematical rigor — General Relativity between 1905 and 1915, quantum mechanics between 1925 and 1932, quantum field theory between the 1930s and the renormalization program of the late 1940s. The Theory of Entropicity (ToE) appears to be in such a stage now: the conceptual architecture is in place, and the program of mathematical concretization is underway.

Looking ahead, Letter III of this series will address the new conservation laws introduced by the Theory of Entropicity (ToE), the Entropic CPT theorem and its implications for symmetry breaking, the Entropic Probability Law, the concept of Self-Referential Entropy, and the implications of the framework for the quantum measurement problem and the black hole information paradox — topics that extend the mathematical architecture reviewed here into new physical domains.

We are, perhaps, at an inflection point in the history of theoretical physics. For a century, the community has sought a unification of quantum mechanics and General Relativity — a theory of quantum gravity that resolves the deep tensions between our two most successful physical theories. The conventional approaches — string theory, loop quantum gravity, causal set theory, asymptotic safety — have made important progress but have not yet achieved the goal. The Theory of Entropicity (ToE) represents a genuinely different approach, one that begins not from geometry or quantum mechanics but from entropy and information. Whether it will succeed is a question that only rigorous mathematics and experimental test can answer. But the question itself is worth taking seriously — and the framework, at the level of its

mathematical architecture, merits the careful, critical, and sustained engagement of the theoretical physics community.

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